

Inverse axiom (2)

$$\begin{array}{ccc} G \cong * \times G & \xrightarrow{e \times \text{id}_G} & G \times G \\ & \searrow \text{id}_G & \downarrow \mu \\ & & G \end{array}$$

$$\begin{array}{ccc} G \cong G \times * & \xrightarrow{\text{id}_G \times e} & G \times G \\ & \searrow \text{id}_G & \downarrow \mu \\ & & G \end{array}$$